

Daily Logic Drill #5

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Flawed Proof Diagnosis · Sign Distribution Traps · Extraneous Solutions · Domain Violations.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Rapid error recognition: algebraic, functional, and logical traps.

1. A student solves $\sqrt{x-2} = -3$ by squaring both sides to obtain $x-2 = 9 \implies x = 11$. Does $x = 11$ satisfy the original equation?
2. A student simplifies $\frac{x^2-4}{x-2} = 4 \implies x+2 = 4 \implies x = 2$. Is $x = 2$ a valid solution to the original equation?
3. In an attempted proof that $1 = 2$, a student starts with $a = b$, obtains $(a-b)(a+b) = b(a-b)$, and concludes $a+b = b$. Which algebraic operation is invalid?
4. A student claims that for all real x , $x^2 > 4 \implies x > 2$. Give a negative integer counterexample to this deduction.
5. A student solves $\log(x) + \log(x-3) = 1$ and finds candidate solutions $x = -2$ and $x = 5$. Which candidate is extraneous?
6. A student conjectures: “If $f'(c) = 0$, then f has a local maximum or minimum at $x = c$.” Give a polynomial function $f(x)$ with $c = 0$ that disproves this claim.
7. A student asserts: “ $|a| < |b| \implies a < b$.” Provide a counterexample pair of integers (a, b) .
8. A student concludes that because n^2 is divisible by 4, n must be divisible by 4. Find the smallest positive integer counterexample n .
9. A student claims: “For all real numbers a, b , if $a < b$ then $a^2 < b^2$.” Give a pair of integers (a, b) with $a < b$ and $a^2 > b^2$.
10. A student asserts: “If $A \subseteq B \cup C$, then $A \subseteq B$ or $A \subseteq C$.” Give a set A of two elements and sets $B = \{1\}, C = \{2\}$ that disprove this claim.

Section B Manipulation Drills

(≤50 s each)

Short numbered proof attempts. Identify what, if anything, is wrong.

- A student attempts to prove: “For all integers n , if n is odd, then n^2 is odd.”

Line 1: Suppose n is odd, so $n = 2k + 1$ for some integer k .

Line 2: Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$.

Line 3: Since $2k^2 + 2k$ is an integer, n^2 is odd.

Is this proof valid? Justify your answer.
- Find the flaw: “Claim: If $x^2 = 3x$, then $x = 3$.” Proof: Divide both sides of $x^2 = 3x$ by x to get $x = 3$.
- Find the flaw: “Claim: The only real solution to $\sqrt{x+7} = x - 5$ is $x = 9$.” Proof: Squaring both sides gives $x + 7 = x^2 - 10x + 25$, so $x^2 - 11x + 18 = 0$, so $(x - 9)(x - 2) = 0$, giving $x = 9$ or $x = 2$.
- A student argues: For every integer n , $6\left(\frac{4n+1}{2} - \frac{n-3}{2}\right)$ is a multiple of 3. (after TMUA 2020 Paper 2 Q3)

Line 1: $6\left(\frac{4n+1}{2} - \frac{n-3}{2}\right) = 3\left(2\left(\frac{4n+1}{2} - \frac{n-3}{2}\right)\right)$.

Line 2: $= 3(4n + 1 - n - 3)$.

Line 3: $= 3(3n - 2)$.

Line 4: which is always a multiple of 3.

On which line does the argument first contain an error, and what exactly is that error?
- Find the flaw: “Claim: For every integer n , $n^2 + n + 2$ is even.” Proof: If n is even, $n = 2k$, so $n^2 + n + 2 = 4k^2 + 2k + 2 = 2(2k^2 + k + 1)$, which is even. Hence $n^2 + n + 2$ is even for every integer n .
- Find the flaw: “Claim: For real x , $x^2 = 9 \iff x = 3$.” Proof: If $x = 3$, then $x^2 = 9$. This proves the biconditional.
- Find the flaw: “Claim: For every positive integer n , the n -th Fibonacci number F_n (with $F_1 = F_2 = 1$) is either equal to 1 or prime.” Proof: $F_3 = 2$, $F_4 = 3$, $F_5 = 5$ — all prime. Since the pattern holds for $n = 3, 4, 5$, the claim is proved. What is the smallest counterexample?
- A student wants to prove: “If n is not divisible by 3, then n^2 is not divisible by 3.” Instead, they show: “If n is divisible by 3, then n^2 is divisible by 3.” Explain why this does not prove the original claim.
- Find the flaw: “Claim: For all integers a, b , if a and b are both even, then $a + b$ is even.” Proof: Since $a + b$ is even, $a + b = 2m$ for some integer m . Since a, b are even, $a = 2j, b = 2k$, so $2j + 2k = 2m$, i.e. $j + k = m$, which is true since m is an integer. Hence $a + b$ is even.
- A student attempts to solve $\log_2(x - 1) + \log_2(x + 1) = 3$:

Line 1: $\log_2((x - 1)(x + 1)) = 3$.

Line 2: $(x - 1)(x + 1) = 2^3 = 8$.

Line 3: $x^2 - 1 = 8 \implies x^2 = 9$.

Line 4: $x = 3$ or $x = -3$.

Line 5: Hence the solution set is $\{-3, 3\}$.

On which line does the first error occur, and what is the true solution set? (after TMUA 2017 Paper 2 Q13)

Section C Substitution & Structure

(≤90 s each)

Tricky MCQs. Some of these proofs are correct — don't assume a flaw exists.

1. Consider the claim: “For every positive integer n , $2n^2 + 2n + 1$ is prime.” (after TMUA 2022 Paper 2 Q7)

Attempted proof:

I. Consider the quadratic $f(x) = 2x^2 + 2x + 1$.

II. The discriminant of $2x^2 + 2x + 1$ is $2^2 - 4(2)(1) = -4 < 0$.

III. Since the discriminant is negative, $2x^2 + 2x + 1$ cannot be factorised into two linear factors with integer coefficients.

IV. Therefore, for every positive integer n , the integer $2n^2 + 2n + 1$ cannot be factorised, so it is prime.

Which of the following best evaluates this argument?

- A) The proof is completely valid, and the claim is true.
- B) The proof is invalid; the error occurs on line II.
- C) The proof is invalid; the error occurs on line III.
- D) The proof is invalid; the error occurs on line IV.
- E) The proof is invalid; the error occurs on line I.
2. Let P : “ $x^2 - 5x + 6 = 0$ ” and Q : “ $x = 2$ ”. A student argues: “Since $Q \implies P$ is true (substituting $x = 2$ gives $4 - 10 + 6 = 0$), and P and Q are ‘clearly related’, we conclude $P \iff Q$.” Which of the following identifies the flaw?
- A) $Q \implies P$ is actually false.
- B) $P \implies Q$ is false; asserting $P \implies Q$ without proof is the error.
- C) There is no flaw; the conclusion is correct.
- D) P and Q are not comparable statements.
- E) The flaw is that Q should be shown to imply $\neg P$.
3. A student attempts to prove: “For every integer n , $n^2 + n$ is divisible by 2.” Proof: Suppose n is even, so $n = 2k$. Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, which is divisible by 2. Hence the claim is proved. Which of the following is the most accurate evaluation?
- A) The proof is complete and correct.
- B) The proof is incomplete: only the even case was considered; the odd case must also be checked.
- C) The proof is invalid because $n = 2k$ is an unjustified assumption.
- D) The claim itself is false; $n = 3$ is a counterexample.
- E) The proof incorrectly divides by 2.

4. A student attempts to prove: “No three consecutive odd positive integers can all be prime.” (after TMUA 2023 Paper 2 Q4)

Line I: Let the three consecutive odd integers be $n - 2, n, n + 2$.

Line II: Exactly one of $n - 2, n, n + 2$ is divisible by 3, for any integer n .

Line III: A positive integer divisible by 3 is composite.

Line IV: So one of the three integers is composite, and therefore no three consecutive odd positive integers can all be prime.

On which line does the argument first break down?

- A) Line I
- B) Line II
- C) Line III
- D) Line IV
- E) The argument is fully correct.

5. A student “proves”: “For every positive integer n , there exists an integer m with $m > n$.” They conclude: “Therefore, there exists an integer M that is greater than every positive integer n .” Which of the following identifies the flaw?

- A) No flaw; both statements are equivalent.
- B) The student has improperly swapped the order of quantifiers.
- C) The flaw is that m must be positive.
- D) The flaw is that no such m exists at all.
- E) The two statements are both false.

6. Claim: “If a quadrilateral is a rhombus, then its diagonals are perpendicular.” A student, wanting to test the converse, argues: “The diagonals of a square are perpendicular, and a square is a rhombus, so this confirms the converse: if a quadrilateral’s diagonals are perpendicular, it is a rhombus.” What is the flaw in this reasoning?

- A) A square is not a rhombus.
- B) One supporting example does not prove a universal converse statement.
- C) The diagonals of a square are not actually perpendicular.
- D) The original claim about rhombi is false.
- E) There is no flaw.

7. A student is asked whether the following claim (*) is true: “For every pair of real numbers x, y , $|x + y| < |x| + |y|$.” (after *TMUA 2020 Paper 2 Q8*)

The student answers: “(*) is false — take $x = -2, y = -5$. Then $|x + y| = |-7| = 7$ and $|x| + |y| = 2 + 5 = 7$, so the required strict inequality $7 < 7$ fails, which disproves (*).”

Assess this answer.

- A) The claim (*) is true, so the disproof must contain a mistake.
- B) A valid counterexample requires x and y to have opposite signs.
- C) A single pair of values is never enough evidence against a “for every” claim.
- D) The student’s reasoning is fully sound.

8. Claim: “Having four equal sides is a necessary and sufficient condition for a quadrilateral to be a square.” A student “verifies” this by checking that every square has four equal sides. What is the flaw in this verification?
- A) Squares do not have four equal sides.
 - B) The student has only checked necessity (square $\implies$ four equal sides), not sufficiency.
 - C) The student has only checked sufficiency, not necessity.
 - D) There is no flaw.
 - E) Both directions were checked correctly.

Section D Challenge

(TMUA / SMC difficulty)

Longer critiques. Some proofs here are entirely valid — read every line before you judge.

1. A theorem and a flawed attempt to prove it: (after TMUA 2021 Paper 2 Q11)
 Theorem: If s, p are real numbers with $s^2 \geq 4p$, then there exist real numbers u, v such that $u + v = s$ and $uv = p$.
 Attempt:
 I. Suppose u, v are real numbers with $u + v = s$ and $uv = p$.
 II. Then $(u - v)^2 \geq 0$, so $u^2 - 2uv + v^2 \geq 0$, i.e. $(u + v)^2 - 4uv \geq 0$.
 III. Substituting, $s^2 - 4p \geq 0$.
 IV. This proves $s^2 \geq 4p$, so the theorem is shown.
 Which of the following best describes this argument?
 - A) It is completely correct.
 - B) It is incorrect; the student proved the converse instead.
 - C) It is incorrect because line II is false.
 - D) It is incorrect because line III is false.
 - E) There is no flaw; assuming u, v exist first is a valid proof technique.

2. A student claims: “For every positive integer n , if $n^2 - n + 11$ is prime, then $n^2 + n + 11$ is also prime.” The student checks $n = 1, \dots, 9$, finds the claim holds in every case, and concludes the claim is proved.
 - (a) Explain why this does not constitute a proof.
 - (b) Find the smallest positive integer n for which the claim fails.

3. A proof of the following claim is given: “For all integers a, b : if ab is even, then a is even or b is even.”
 Proof attempt: Suppose, for contradiction, that ab is even but a and b are both odd. Write $a = 2j + 1, b = 2k + 1$. Then $ab = 4jk + 2j + 2k + 1 = 2(2jk + j + k) + 1$, which is odd. This contradicts ab being even. Hence a is even or b is even.
 Which of the following statements about this proof is/are true?
 - I. The proof is a valid proof by contradiction.
 - II. The proof could equally well be presented as a direct proof by contrapositive.
 - III. The proof contains an algebraic error in expanding $(2j + 1)(2k + 1)$.
 - A) I only
 - B) II only

- C) III only
- D) I and II only
- E) I, II and III
- F) none

4. The following argument is given for the claim “every odd prime greater than 3 can be written as $6k \pm 1$ for some positive integer k ”:

Line 1: Every integer can be written as $6k, 6k + 1, 6k + 2, 6k + 3, 6k + 4$, or $6k + 5$ for some integer $k \geq 0$.

Line 2: $6k, 6k + 2, 6k + 4$ are all even, so none can be an odd prime greater than 3.

Line 3: $6k + 3 = 3(2k + 1)$ is divisible by 3, so cannot be a prime greater than 3.

Line 4: This leaves only $6k + 1$ and $6k + 5 = 6(k + 1) - 1$, i.e. of the form $6k' - 1$. So every odd prime greater than 3 has the form $6k \pm 1$.

Which of the following is true?

- A) The argument is fully valid, and correctly establishes the claim exactly as stated.
- B) The argument proves necessity but fails to prove sufficiency.
- C) There is a genuine gap: line 2 incorrectly dismisses $6k + 4$.
- D) There is a genuine gap: line 3 is false.
- E) The proof is invalid because it never explicitly rules out $6k$ itself.

5. A student attempts to prove: “For all positive integers $n \geq 2$, $n^2 - 1$ is composite.”

Line 1: $n^2 - 1 = (n - 1)(n + 1)$.

Line 2: For $n \geq 2$, $n - 1 \geq 1$ and $n + 1 \geq 3$.

Line 3: Since $n^2 - 1$ is written as a product of two integers each at least 1, it is composite.

Line 4: Hence $n^2 - 1$ is composite for all $n \geq 2$.

(a) Identify the line containing the first logical gap.

(b) Give the smallest $n \geq 2$ for which the conclusion in Line 4 is actually false, and prove that this is the *only* value of $n \geq 2$ for which it fails — i.e. that the conclusion is genuinely correct for every $n \geq 3$.

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths